Humanoid Social Navigation: State of the Art, Evidence Limitations, and Research Opportunities for Geoff-Sim

Project literature review and research-positioning document

Literature search and repository inspection: 22 September 2026

Abstract

Geoff-Sim integrates Unitree G1 locomotion, MuJoCo, ROS 2, online mapping, pedestrian tracking, human-aware planning, collision supervision, and experimental benchmarking. This review evaluates 50 identity-verified research publications and preprints spanning 1995–2026. It connects classical collision avoidance, learning-based crowd navigation, trajectory prediction, legged control, onboard perception, and social-navigation evaluation. Recent work already demonstrates LiDAR-conditioned humanoid avoidance, cross-embodiment navigation, and physically simulated multi-task benchmarks. Consequently, the strongest positioning is not the first integration of walking and social navigation. It is a falsifiable investigation of how calibrated locomotion response, observation age, uncertainty, and planning/supervision interaction affect closed-loop collision risk and mission liveness on a walking humanoid. The repository has unusually useful information-source and experiment-validity instrumentation, but its documented onboard trials still include collisions and timeouts. Five research opportunities are ranked, with explicit baselines, ablations, metrics, and claim boundaries. Source identity verification is separated from abstract-level screening, selected full-text analysis, and experimental replication; the latter was not undertaken.

Contents

1 Executive assessment **3**

2 Scope, search procedure, and verification standard **3**

2.1 Review type and research questions 3

2.2 Search and inclusion procedure 4

2.3 What “verified” means here 4

3 The current work and its evidence boundary **5**

3.1 Architecture relevant to the review 5

3.2 Oracle and sensor-only modes are different experiments 5

3.3 What the recorded trials establish 5

3.4 The key technical opening 5

4 Classical and learning-based social navigation **6**

4.1 Geometric and dynamically constrained avoidance 6

4.2 Interaction, reciprocity, and freezing 6

4.3 Learned interaction models 7

5 Prediction and onboard perception **7**

5.1 From point forecasts to interaction-conditioned distributions 7

5.2 Detection and tracking are not solved by geometry alone 8

5.3 Localization, mapping, and the moving sensor 8

6 Legged control and the closest humanoid competitors **9**

6.1 Locomotion competence versus navigation competence 9

6.2 ASAP: why nominal simulator dynamics are not enough 9

6.3 LiDAR-conditioned safe and comfortable humanoid locomotion 10

6.4 FLUX: a 2026 cross-embodiment navigation competitor 10

7 Safety, middleware, and whole-system timing **11**

7.1 Filtering commands versus guaranteeing safety 11

7.2 Planner–supervisor interaction 11

7.3 ROS 2, Nav2, and SLAM are enabling infrastructure 12

8 Benchmarking and social validity **12**

8.1 Existing benchmarks already cover important parts of the problem 12

8.2 Safety is necessary but not sufficient for social navigation 12

8.3 A useful validity hierarchy 13

9 Comparative synthesis and ranked research gaps **13**

9.1 Gap 1: calibrated executable avoidance under stale observations 14

9.2 Gap 2: causal explanation of the oracle-to-onboard gap 14

9.3 Gap 3: coordinated safety and liveness across planning layers 14

9.4 Gap 4: full-body risk beyond planar clearance 15

9.5 Gap 5: contextual comfort and nonreciprocal crowds 15

10 A concrete experimental programme **15**

10.1 Stage A: characterize the frozen locomotion interface 15

10.2 Stage B: controlled oracle planning study 16

10.3 Stage C: factorial onboard robustness study 16

10.4 Stage D: metrics and statistical interpretation 16

10.5 Minimum publishable evidence package 17

11 Recommended positioning and conclusion **17**

A Annotated evidence map of the 50 retained works **17**

B Claim audit and immediate priorities **19**

1 Executive assessment

Recommended core research question: *Can a calibrated, observation-age-aware model of a humanoid's executable motion improve the collision–completion trade-off of social navigation under partial observability, relative to matched geometric, temporal, and fixed-margin baselines?*

This question fits the current implementation closely. It makes the robot's actual delayed response, rather than its requested planar velocity, the object of prediction. It also recognizes that a planner using a fresh message can still be acting on an old observation. Its novelty must be established by the specific model, calibration procedure, decision rule, and controlled evidence, because dynamics-aware avoidance and biped crowd navigation are established topics [6, 15, 16].

Three conclusions organize the review:

1. **The field has moved beyond isolated obstacle avoidance.** Recent systems combine learned locomotion with perception and navigation; the closest humanoid paper maps LiDAR and proprioception directly to motor commands, and FLUX reports transfer to a physical G1 [18, 48]. A general claim that previous research ignores humanoids, dynamics, or raw sensing is untenable.
2. **The remaining problem is measurable closed-loop reliability.** Forecast accuracy, nominal command feasibility, or a visually successful demonstration does not establish low collision risk over complete missions. Perception loss, stale observations, tracking association, command shaping, and stopping transients can interact. Current repository documentation provides concrete examples of this gap.
3. **Evaluation can be a substantive contribution if it tests a new scientific question.** Separating oracle control, sensor-only control, and independent ground-truth scoring enables causal diagnosis. The contribution would be a validated protocol and new findings about failure mechanisms, not simply another simulator or another collection of ROS nodes.

The most defensible near-term paper is a simulation-based methods and evaluation study. Hardware transfer and human comfort should be treated as subsequent validation targets, not inferred from MuJoCo clearance measurements.

2 Scope, search procedure, and verification standard

2.1 Review type and research questions

This is a **structured critical scoping review**, not an exhaustive systematic review or a meta-analysis. It asks:

1. What is already established about planning around people while respecting the dynamics and sensing constraints of a mobile or legged robot?
2. Which recent humanoid and cross-embodiment systems are direct competitors?
3. What evidence distinguishes geometric avoidance, social acceptability, and reliable mission-level autonomy?
4. Which gaps can this repository realistically investigate, and what would falsify the proposed advantages?

The search cutoff is 22 September 2026. Foundational work is retained when it defines a necessary baseline or assumption. The latest included preprint, DPed-VLN, was submitted on 18 September 2026 [14]; its inclusion is an abstract-level horizon scan, not a claim of mature validation. Date and venue are taken from retrieved records, not inferred from identifier prefixes.

2.2 Search and inclusion procedure

Discovery used Crossref title/bibliographic searches, arXiv title and topic searches, and references encountered in closely related papers. OpenAlex was used as an auxiliary locator for an accessible copy of the older biped paper. Representative topic strings were "social humanoid navigation", "socially aware robot navigation", "safe navigation biped crowd", "LiDAR legged collision avoidance", "humanoid sim-to-real", and exact method titles. The reproducible query families and screening decisions are recorded in `SEARCH_LOG.md`. Exact selected identifiers are in `sources.json`.

Inclusion required a direct connection to at least one of: human-aware navigation; crowd perception/prediction; executable collision avoidance; legged/humanoid control; simulation/middleware used by the project; or evaluation methodology. Foundational infrastructure papers are included to separate enabling technology from methodological novelty. Generic language models, manipulation-only systems, and unrelated humanoid demonstrations were not used to inflate the count. Title-search mismatches and duplicate versions were rejected. For example, a search for FAST-LIO2 returned both an SSRN record and the IEEE article; the IEEE record was selected.

The retained corpus contains **50 distinct works: 41 Crossref DOI records and nine arXiv records**. An arXiv citation does not imply that the work is unpublished; it means the cited version was verified through arXiv rather than an independently checked proceedings record. Publication-year differences between preprints, early access, and journal issues are normal. This review uses the year returned for the selected record.

2.3 What "verified" means here

Verification has three distinct levels:

Identity verification (all 50). A primary bibliographic record was retrieved from Crossref or arXiv. Title, author list, year, DOI/arXiv identifier, and source URL were extracted and checked. The evidence snapshot records the retrieval timestamp and SHA-256 of the response. This establishes that a matching paper record exists; it does not establish the truth of its findings.

Content inspection. Registry abstracts and additional arXiv abstracts were inspected where available. The closest recent competitors—ASAP, Safe and Comfortable Locomotion, FLUX, and OmniNavBench—received selected full-text methods/evaluation analysis. Specific table-level criticisms below refer to those inspected versions. Other works primarily supply conceptual context and candidate baselines; uninspected numerical claims are avoided.

Replication. No external algorithm was reimplemented or rerun for this review. No new robot experiment or human-subject evaluation was performed. Repository results are explicitly attributed to existing development records.

`verified_sources.json` retains the identity evidence and available abstracts. `READING_AUDIT.md` identifies the inspected material and remaining access limitations. `verify_sources.py` checks the local snapshot, bibliography, and citation keys offline, and can refresh the online records. Full-text access was uneven: the HAL copy of the 2016 biped paper presented an access challenge, so this review does not attribute detailed experimental claims to it. Search ranking, indexing delays, language coverage, and a single-pass synthesis limit completeness. A final publication should extend forward/backward citation screening and independently review its novelty claim.

3 The current work and its evidence boundary

3.1 Architecture relevant to the review

Geoff-Sim is a simulation-first Unitree G1 research platform. Its documented pipeline connects MuJoCo/DDS robot state and simulated sensors to ROS 2, SLAM Toolbox, Nav2, dynamic-person tracking, short-horizon prediction, human-aware command selection, a collision supervisor, and a learned locomotion controller. This is an appropriate architecture for investigating the interfaces between autonomy layers, rather than only isolated planner performance.

The repository inspection covered `README.md`, `docs/ONBOARD_SOCIAL_NAVIGATION.md`, `docs/DYNAMIC_ROUTE_PLANNING.md`, `docs/BENCHMARK_EXPERIMENTS.md`, and the implementation `motion_envelope.py` in the bridge package. File hashes and the repository revision/dirty-state context are retained in `project_evidence.json`. The working tree contained ongoing development; the reviewed files, not the base commit alone, define this assessment's project snapshot.

3.2 Oracle and sensor-only modes are different experiments

The oracle social-waypoint mode uses simulator-derived person tracks and a known static map. The onboard mode uses local-frame raycast LiDAR, IMU and joint information, scan-registration odometry, online SLAM, observation-based tracking, and constant-velocity pedestrian forecasts. Evaluation truth remains available for scoring but is excluded from the documented navigation path. The source audit is a graph/configuration check rather than a security sandbox. This distinction is scientifically important: privilege can enter through localization, goal selection, maps, or tracks, not only through a planner's direct subscriptions.

The onboard detector is geometric, not a semantic camera or learned person classifier. The odometry front end assumes approximately planar walking, and its covariance is not a calibrated localization uncertainty model. Consequently, the present system provides an experimental platform for perception-aware navigation, rather than a completed solution to open-world person recognition or general uneven-terrain localization.

3.3 What the recorded trials establish

The onboard status document explicitly states that an earlier oracle 24/24 result has not been reliably reproduced with onboard sensing. Development trials include an eight-goal completion and several collisions and timeouts. These runs span implementation changes and must not be pooled into a single algorithm success rate. The dynamic-route validation records 136 route updates but only three of four goals and one collision. This establishes active integration, not reliable avoidance.

Timing records are similarly informative but conditional. One onboard trial reports planner p95 47.115 ms, tracker p95 53.501 ms, and track age at output p95 0.305 s. A later route-validation trial reports request-to-matching-route p95 2.649 s, including retries and blocked-route periods. These quantities measure different things: node computation time, observation age, and delivered route latency cannot be substituted for one another. None is a hard real-time guarantee. The reported 45 ms candidate-search budget can overrun.

3.4 The key technical opening

The shared motion envelope already normalizes command bounds and models acceleration-limited, delayed planar motion. Its source explicitly states that the defaults are *conservative en-*

gineering assumptions, not a fitted G1 response certificate. That admission identifies a tractable research step: measure the command-to-motion relationship and test whether a calibrated model changes safety decisions. Current velocity caps vary by profile, so experiments must freeze the policy, gains, normalization, delay model, and active command limits instead of quoting one cap as universal.

The batch tools already preserve failed/invalid/missing trials and distinguish control-source evidence from scoring truth. These are useful foundations, but their presence does not by itself validate the response model or prove a novel navigation method.

4 Classical and learning-based social navigation

4.1 Geometric and dynamically constrained avoidance

The Dynamic Window Approach (DWA) searches feasible velocities subject to the robot's dynamics and stopping considerations [16]. Velocity obstacles (VO) reason about relative motion and select velocities outside collision-producing regions; the original work also intersects these with acceleration-admissible velocities [15]. Therefore, merely adding an acceleration limit to a geometric planner is not a sufficient novelty claim. Their continuing value is as interpretable baselines: failures can be traced to motion estimates, admissible sets, or objective functions.

Timed trajectory optimization offers another classical comparator. Sparse trajectory optimization underlying the timed-elastic-band family jointly adjusts a path and its temporal structure [38]. It makes time and kinematic constraints explicit, but practical behavior still depends on initialization, obstacle representations, and the fidelity of its executable motion model. A G1 can have command deadbands, axis coupling, turning-arc restrictions, and balance-related transients that are not captured by a generic planar robot model.

ORCA distributes collision-avoidance responsibility between interacting agents [46]. This is powerful when agents conform sufficiently to the reciprocal assumptions. The repository's independently walking actors do not react to the robot, so an unmodified reciprocal assumption can be misleading. An ORCA comparison should declare whether people participate in the same avoidance rule, remain nonreactive, or follow recorded trajectories. A planner baseline and a pedestrian simulator are different uses of ORCA and should not be conflated.

Research implication. The useful distinction is not "dynamics-aware versus all previous kinematic work". It is whether a particular model of the deployed locomotion policy, command filtering, and observation delay improves decisions beyond established feasible-velocity and timed-trajectory methods.

4.2 Interaction, reciprocity, and freezing

Social-force modeling provides a compact account of interactions among pedestrians [21]. Its interpretability makes it useful for simulation and baseline reasoning. It does not make a simulated crowd a validated model of all human behavior, particularly for groups, distraction, or unexpected encounters with a humanoid. Crowd generation should therefore be an experimental factor rather than an unquestioned ground truth.

Trautman and colleagues explicitly study cooperation in dense crowds [45]. The central conceptual challenge is that treating every possible human motion as an obstacle can prevent progress, while assuming cooperation can be unsafe when the assumption is wrong. The same tension appears in Geoff-Sim: stronger margins may prevent a local encounter but produce waypoint timeouts; a rapid command can improve progress while increasing collision or fall risk.

Performance should be evaluated on a collision–completion frontier rather than a single "safety score".

The early human-aware-navigation survey and subsequent taxonomies emphasize that obstacle avoidance is only one part of navigation among people [25, 42]. A robot must also make its behavior legible, respect interaction context, and avoid disrupting people. A successful nonreactive-actor benchmark validates unilateral avoidance under a defined motion model. It does not measure cooperative negotiation or human comfort.

4.3 Learned interaction models

CADRL learns a value function for decentralized collision avoidance, moving much of the interaction computation into offline training [12]. SARL uses attention to reason about collective human interactions rather than only independent robot–human pairs [10]. Relational Graph Learning adds graph-structured representations and predictive lookahead [11]. DS-RNN incorporates spatial and temporal relationships and addresses partial observability; its abstract also reports transfer to a TurtleBot 2i [28]. These methods provide a useful progression from pairwise local reasoning to temporal crowd representations.

The critical comparison is the information and dynamics contract. A policy consuming tracked positions and velocities does not perform the detector's task. A benchmark using a circular planar robot does not necessarily test balance, upper-body swing, or actual stopping transients. Conversely, it would be inaccurate to label all learning-based methods "oracle-only" or "simulation-only", particularly given DS-RNN and newer sensor-level systems. Each comparator needs its own documented observation interface.

The 2025 review and benchmarking paper identifies strengths of learning-based and end-to-end approaches in coordination scenarios [2]. This supports including a learned navigation comparator if resources permit. However, importing a policy trained with different pedestrian reactions, speed ranges, sensors, or reward conventions can create a distribution-shift test rather than a fair algorithm comparison. Such transfer is valuable, but should be labeled and analyzed separately from matched training.

Research implication. A modular response-aware approach is attractive for interpretability and data efficiency, not because learned policies cannot avoid people. Its case depends on comparable input information and a demonstrated advantage under delay, missed detections, or changed locomotion response.

5 Prediction and onboard perception

5.1 From point forecasts to interaction-conditioned distributions

Social LSTM introduced learned interaction-aware trajectory prediction; Social GAN explicitly represents diverse socially plausible futures; Trajectron++ combines graph structure, heterogeneous agent data, and dynamical feasibility [1, 19, 41]. Trajectron++ can condition predictions on an ego-agent plan, which is especially relevant when robot action influences pedestrian behavior. Rudenko et al.'s survey provides a broad taxonomy of trajectory modeling and contextual information [39].

These developments challenge a single constant-velocity forecast over long horizons, but do not imply that a larger network will improve this project's closed-loop behavior. Short-horizon constant velocity is a valuable baseline: it is inexpensive, auditable, and can be more stable than acceleration extrapolation from noisy surface centroids. The onboard profile's use of constant

velocity is therefore defensible as a starting point. Its limits must be measured at crossings, turns, stops, and occlusion exits.

Prediction error averaged over entire trajectories can conceal the errors that cause a collision. Multi-sample best-of-K displacement error can also reward producing one good sample without providing calibrated probabilities. For planning, the relevant questions are whether the occupied region covers the person at encounter time, whether probability estimates are calibrated, and whether the chosen robot motion remains feasible. This is a synthesis of the planning problem, not a claim that all reviewed predictors omit these issues.

For a track observed at t_o and used at t_c, define its age $a = t_c - t_o$. A minimal time-aligned mean prediction is

$$\hat{p}_h(t_c + \tau) = \hat{p}_h(t_o) + \hat{v}_h(t_o)(a + \tau). \tag{1}$$

The corresponding occupied region must also widen with measurement, velocity, and unmodeled-motion uncertainty. Equation 1 is only a proposed baseline formulation; it is not evidence that the current implementation has calibrated coverage. Repeated publication of an old track must not reset t_o.

5.2 Detection and tracking are not solved by geometry alone

DROW establishes learned detection from sparse two-dimensional range data, initially emphasizing wheelchairs [4]. DR-SPAAM uses spatial attention and recurrent feature updates for person detection in range scans [23]. These are pertinent perception baselines, but transferring their assumptions to a moving torso-mounted 3D LiDAR requires an explicit projection and domain-adaptation study. A flat scan through legs is not equivalent to clipped, oscillating torso returns.

JRDB provides egocentric human detection/tracking data in built environments [32]. SCAND provides human-demonstrated socially compliant navigation on morphologically different platforms, including a quadruped and a wheeled robot [24]. The former is useful for characterizing missed detections and association; the latter is useful for behavioral imitation and judging which maneuvers humans demonstrate. Neither automatically supplies the G1-specific LiDAR mounting, sway pattern, or command response of this project.

The repository already attempts partial-body track retention and measured fragment recovery. These are practical responses to a genuine problem: an observed surface centroid is not a stable estimate of a person's body center. Differentiating that centroid may produce an apparent acceleration that belongs to visibility geometry rather than human motion. Ground-truth evaluation should separate center bias, velocity error, identity continuity, track age, and complete misses. The current nearest-track diagnostic is explicitly not identity-matched detection recall; it should not be reported as such.

5.3 Localization, mapping, and the moving sensor

FAST-LIO2 demonstrates direct LiDAR–inertial estimation, while ORB-SLAM3 provides a visual/visual–inertial alternative [8, 50]. They identify strong adjacent baselines and underline that state estimation is a developed field. The project's planar scan-registration front end should be positioned as an engineering component unless a new estimation method is separately evaluated.

Walking creates a feedback loop between body motion, sensing, and planning. Attitude errors can affect ground removal and scan projection; localization drift can shift both maps and tracks; dynamic-person masking can remove useful geometry. Correctly handling unknown map

cells does not by itself resolve these effects. Conversely, adding a more capable estimator may not fix collision failures caused by stale tracks or unmodeled braking.

Research implication. A controlled oracle-to-onboard decomposition is stronger than a single end-to-end success number. Independently substitute estimated pose, observed tracks, and online maps to locate which uncertainty source changes decisions. Then test interactions between those sources rather than assuming their effects add linearly.

6 Legged control and the closest humanoid competitors

6.1 Locomotion competence versus navigation competence

Massively parallel training reduces the time needed to learn locomotion policies [40]. RMA learns rapid adaptation to changing locomotion conditions [26]. Robust perceptive locomotion combines exteroception and proprioception to handle difficult terrain [33]. These advances make a learned low-level policy an increasingly practical component, but their task objectives differ from mission-level navigation through a crowd. Good velocity tracking in a locomotion benchmark does not specify a safe pedestrian-avoidance trajectory.

DeepMimic establishes example-guided physics-based motion imitation [35]; ExBody transfers expressive whole-body behavior to humanoids while relaxing lower-body imitation to support velocity tracking [13]. These are relevant to natural motion and potentially legibility, but motion resemblance is not evidence of perceived social comfort. Human evaluation would be needed to establish that relationship.

The boundary between locomotion and navigation is already an active research area. ANYmal Parkour uses a hierarchy that accounts for locomotion skills when choosing how to traverse obstacles [22]. The wheeled-legged navigation work integrates adaptive locomotion, mobility-aware local navigation, and large-scale planning, reporting kilometer-scale missions [27]. Earlier work explicitly addresses safe biped navigation in crowds [6]. These are direct counterexamples to an overly broad claim that navigation has always ignored legged execution.

6.2 ASAP: why nominal simulator dynamics are not enough

ASAP pretrains motion-tracking policies, collects real-world trajectories, learns a residual action model to align simulated response, and fine-tunes the policy in the corrected simulator [20]. The inspected version compares transfer between simulators and to a real G1. Its tables examine both open-loop dynamics mismatch and closed-loop motion imitation. This separation is important: fitting transitions and improving a deployed policy are related but distinct achievements.

For Geoff-Sim, ASAP strengthens the motivation to calibrate response rather than trust nominal gains. It also limits the novelty of "learning dynamics corrections". The proposed contribution here would operate at a different interface: a short-horizon reachable motion envelope for a frozen locomotion policy, used to select or veto navigation commands. It would not duplicate ASAP's whole-body policy fine-tuning objective. Dynamics randomization is another established transfer tool [36], but randomized training does not supply a calibrated deployment-time stopping bound by itself.

ASAP is not a crowd-navigation evaluation. Its imitation errors and skill success rates cannot be numerically compared with pedestrian collision or waypoint completion rates. The useful transfer is methodological: evaluate model fidelity and downstream behavior separately, on held-out conditions.

6.3 LiDAR-conditioned safe and comfortable humanoid locomotion

The most directly overlapping 2025 work is *End-to-End Humanoid Robot Safe and Comfortable Locomotion Policy* [48]. It combines LiDAR features and proprioceptive history, uses a constrained Markov decision process, translates discrete barrier-function ideas into training costs, and incorporates comfort-oriented rewards. The inspected paper reports Unitree G1/Livox Mid-360 deployment and onboard execution. Its actor/critic description explicitly distinguishes privileged training information from deployment inputs.

This prior work makes "first sensor-driven safe humanoid avoidance" an indefensible positioning. It also offers a useful contrast between an end-to-end learned controller and a modular planner plus supervisor. However, several details constrain how strongly its results can be used:

- Table III reports 30 trials per scenario and rounded success values of 86% for the proposed method versus 70% for P3O in the dynamic-agent case. The same proposed method reports 60% in a narrow passage. These are scenario-dependent outcomes, not universal avoidance guarantees.
- Table II reports nonzero unsafe-region time even for the proposed method. The training discussion uses a 0.8 m safety distance, while that table defines unsafe space below 0.6 m. The evaluation thresholds must be kept separate rather than merged into a single certified clearance claim.
- The described barrier condition is converted into an optimization cost during learning, rather than enforced as an online action constraint. A theoretical invariant-set condition is not automatically inherited by an approximately trained policy under sensing and dynamics errors.
- Comfort is operationalized through reward terms and geometric/dynamic proxies. The inspected experiments do not establish perceived comfort through a reported participant study. This leaves room for later HRI validation, but not a claim that comfort was wholly absent from prior work.

These observations are not a re-evaluation of the paper's code or raw data. They identify which claims its printed evidence can support and which quantities would need harmonization in a comparative experiment.

Omni-Perception is another close 2025 comparator: its abstract describes raw LiDAR, spatiotemporal risk processing, omnidirectional avoidance, and a simulation toolkit compatible with multiple engines [47]. It prevents novelty from being assigned merely to direct point-cloud perception or multi-simulator raycasting. Its detailed comparative results were not independently inspected in this review.

6.4 FLUX: a 2026 cross-embodiment navigation competitor

FLUX uses rectified-flow trajectory generation followed by dynamic-scene reinforcement-learning refinement, and introduces DynBench [18]. The inspected version includes static and dynamic navigation tasks and reports transfer to wheeled, quadrupedal, and G1 humanoid platforms. Its deployment uses an RGB-D camera and inference on a remote RTX 4060 machine. This matters when comparing onboard compute constraints and communication latency with Geoff-Sim.

Three details are especially relevant to critical positioning:

1. Its Table II reports SocialNav success of 64.0 versus 59.3 for NavDP. This is meaningful within that protocol, but goals, episode termination, pedestrian speeds, and social-distance definitions differ from Geoff-Sim. It is not a directly comparable headline improvement.
2. Section V defines "Collision Count" as intimate-space intrusions below 0.45 m, a proxy for collision risk. Later prose describes values with percent signs while the metric definition

describes counts. Accordingly this review does not reinterpret those values as physical contact probabilities. Physical contacts and proxemic intrusions should be distinct outputs in the proposed study.

3. Table IV reports 216.5 ms inference for the selected six-step configuration. This is a different timing boundary and compute setup from the project's local-planner p95 measurements. It motivates measuring end-to-end observation-to-actuation age; it does not prove one system is faster under matched conditions.

FLUX's limitation discussion acknowledges missing explicit social reasoning and predominantly simulated indoor pedestrian evaluations. Its existence still decisively shifts the research gap: cross-embodiment dynamic navigation is no longer an untouched area. A narrowly scoped G1 response and uncertainty study can be complementary to such general policies.

7 Safety, middleware, and whole-system timing

7.1 Filtering commands versus guaranteeing safety

Control barrier functions provide a formal route to invariant-set conditions and quadratic-program safety filtering under stated model assumptions [3]. The safe-learning survey distinguishes empirical robustness, learning-based improvement, and formal certification [7]. These distinctions should govern terminology in this project. A collision supervisor that checks predicted rollouts is a safety-oriented mechanism; without validated disturbance bounds and recursive feasibility, it is not automatically a safety certificate.

A navigation command u is transformed before its physical effect appears. Let $\mathcal{N}$ denote normalization, deadbands, and command shaping, and let $\hat{f}_\theta$ approximate policy-level motion response. A candidate rollout can be expressed as

$$\hat{x}_{k+1} = \hat{f}_\theta(\hat{x}_k, \mathcal{N}(u_k), d_k), \tag{2}$$

where d_k includes the measured or modeled delay. This notation summarizes a proposed experimental model, not a claimed full-order description of G1. Its accuracy must be measured during braking, turning, and combined-axis motion, not only steady straight walking.

For a one-dimensional stop with constant speed during delay and constant deceleration afterward, an illustrative distance is

$$d_{\text{stop}} = vd + \frac{v^2}{2a_{\text{brake}}}, \qquad a_{\text{brake}} > 0. \tag{3}$$

The equation explains why zero commanded velocity is not zero stopping distance. It is not a bound until delay and achievable braking have been validated, and it omits a moving pedestrian's continuing approach. For example, if a person approaches at 1 m/s, an observation age of 0.305 s corresponds to 0.305 m of unobserved motion before future-horizon prediction even begins. This is an illustrative calculation using a chosen human speed, not a reconstructed collision from the repository.

7.2 Planner–supervisor interaction

A planner may choose a trajectory that the supervisor repeatedly vetoes. Even when every individual veto seems prudent, repeated disagreement can cause oscillation or starvation. A suitable study measures override rate, rejected-command reasons, clearance under the actual applied command, yield duration, and progress after resumption. It should also distinguish

stopping in front of an approaching person from moving to a genuinely lower-risk configuration: stationary is not synonymous with safe in a dynamic environment.

For a humanoid, emergency velocity changes can create a second hazard: instability or contact by a swinging body segment. A planar footprint can be useful if conservatively validated, but it does not directly measure full-body swept volume, fall reach, or contact force. Such distinctions are particularly relevant to the project's use of independent pedestrian actors.

7.3 ROS 2, Nav2, and SLAM are enabling infrastructure

ROS 2 provides the communication and lifecycle architecture [31]. Navigation2's behavior-tree orchestration is described in The Marathon 2 [30], and SLAM Toolbox supports the mapping component [29]. MuJoCo provides the physics foundation [44]. Combining these tools is valuable research engineering, but their presence is not a new navigation algorithm. Nor does middleware design establish hard real-time performance for an entire application.

The current event-driven route monitor, forecast costmap layer, and retained path execution offer a specific timing experiment: when does event-triggered replanning improve collision-free progress relative to fixed-rate replanning under the same compute budget? A 5 Hz request limit is not a 200 ms route delivery guarantee. Queueing, costmap availability, retries, and action transport belong in the latency measurement. Global predictive costs and local collision checking should also be ablated independently to avoid crediting the same forecast twice.

8 Benchmarking and social validity

8.1 Existing benchmarks already cover important parts of the problem

SocNavBench provides grounded scenarios, consistent metrics, and a common testing framework [5]. SocialGym 2.0 emphasizes configurable multi-robot navigation in shared human spaces [9]. Habitat 3.0 adds human avatars, human-in-the-loop interaction, and collaborative tasks [37]. MetaUrban extends compositional urban environments and heterogeneous mobility evaluation [49]. These address realism, comparability, interaction, or scale that a small custom plaza cannot assume.

OmniNavBench is particularly important to a 2026 gap analysis [43]. Its inspected version includes 170 environments and 1,779 human-teleoperated trajectories, multiple embodiments including Unitree H1, continuous physical stepping, and composed navigation tasks. The low complete-task success of several evaluated models highlights a distinction between individual skills and reliable task sequences. Its social metric uses a personal-space threshold, which should not be equated with a full-body contact sensor. Geoff-Sim's proposed advantage should therefore be fine-grained control/measurement provenance and calibrated execution analysis, rather than the first physically embodied multi-task benchmark.

The very recent DPed-VLN abstract describes language navigation with dynamic pedestrians, paired instructions, and social-safety metrics [14]. It is adjacent rather than a direct baseline for a LiDAR/Nav2 waypoint stack. Its inclusion demonstrates why "navigation in dynamic pedestrian scenes" is too broad a novelty claim. This review does not infer its low-level humanoid dynamics or deployment properties from the presence of humanoid pedestrian avatars.

8.2 Safety is necessary but not sufficient for social navigation

The evaluation guidelines articulate safety, comfort, legibility, politeness, social competency, agent understanding, proactivity, and context responsiveness [17]. The 2026 proxemics taxonomy

further emphasizes human, robot, environment, and contextual influences [34]. A fixed circular personal-space penalty is consequently a controllable proxy, not a universal measure of comfort.

A simulator can validly measure minimum body clearance, contact events, time in specified proxemic regions, approach speed, jerk, and route progress. To infer perceived safety or comfort, the study needs appropriately designed human judgments, such as counterbalanced video ratings or later controlled in-person comparisons. These should separate morphology, speed, yielding style, and context. A humanoid's natural-looking gait may improve legibility or may increase expectations that the controller does not meet; the direction of the effect is an empirical question.

8.3 A useful validity hierarchy

1. **Mock regression:** verifies scenario, scoring, aggregation, and failure-handling mechanics; it provides no physical avoidance evidence.
2. **Oracle physics:** tests planning and execution with privileged perception; it cannot establish sensor-only performance.
3. **Onboard-sensor simulation:** tests the integrated observation and execution pipeline under the simulator's sensing/physics assumptions.
4. **Cross-simulator or perturbed-model evaluation:** probes sensitivity to selected modeling assumptions; it is not itself hardware transfer.
5. **Hardware and human evaluation:** addresses physical transfer and human-centered outcomes under an explicit deployment protocol.

The current project spans the first three levels with incomplete reliability at level three. Reporting the hierarchy avoids overstating either the mock tests or the passing oracle demonstrations.

9 Comparative synthesis and ranked research gaps

Table 1: Closest families and the remaining project-specific question. This is a qualitative synthesis, not a benchmark ranking.

Prior work	Established capability	Question still testable here
DWA, VO, TEB, biped crowd navigation	Dynamics/feasibility matter in avoidance	Does a calibrated policy-level response envelope outperform generic bounds?
CADRL, SARL, RGL, DS-RNN	Learned spatial/temporal crowd reasoning	What happens under matched sensing age, misses, and G1 execution?
ASAP, RMA, perceptive locomotion	Adaptation and dynamics-aware learned control	Can frozen-policy response calibration improve navigation without retraining locomotion?
Safe humanoid policy, Omni-Perception	Sensor-conditioned legged/humanoid avoidance	How do modular filtering and end-to-end policies trade contact risk against liveness?
FLUX, OmniNavBench	Cross-embodiment navigation and richer evaluation	How much apparent performance changes with information provenance and measured execution delay?
Social-navigation guidelines and proxemics taxonomy	Multidimensional human-centered evaluation	Which safety/progress gains correspond to judged comfort and legibility?

9.1 Gap 1: calibrated executable avoidance under stale observations

Priority: highest; fit: direct; novelty confidence: moderate and conditional. Established feasible-velocity methods, biped planning, and recent LiDAR policies already address parts of the problem [6, 15, 16, 48]. The narrower unresolved question for this platform is the contribution of a *measured policy-level response distribution* combined with observation-age-aware person occupancy.

Hypothesis H1. At matched clearance conservatism and compute budget, a calibrated response envelope lowers contact incidence or increases collision-free mission completion compared with instantaneous and hand-tuned response models.

Contribution required. Define what is calibrated: delay, gain, acceleration/braking asymmetry, yaw response, combined-axis coupling, and residual prediction error. Condition or stratify by relevant initial motion. Use held-out commands and scenes, and propagate residual uncertainty into the navigation decision. The estimator and its uncertainty treatment must be specified sufficiently to reproduce.

Falsification. If an equally conservative fixed-margin baseline matches the result, or improvements disappear on held-out response tests, the extra model complexity is not justified. A finding that timing dominates model complexity would still be useful if established through controlled interventions.

9.2 Gap 2: causal explanation of the oracle-to-onboard gap

Priority: high; fit: direct; novelty confidence: moderate. Benchmarks and perception datasets already exist [5, 32, 43]. What can be distinctive is a source-audited experimental decomposition on the *same walking system*.

Hypothesis H2. During close encounters, track age and partial-body velocity bias explain more avoidable failures than map or localization error alone, after controlling for scenario and locomotion policy. This is a hypothesis suggested by the repository's diagnostics, not an established causal result.

Contribution required. Independently replace observed person tracks, estimated robot pose, and online map with their oracle counterparts. Keep goal definitions, actor motion, policy, and evaluation truth fixed wherever possible. Then inject controlled delay, track dropout, identity changes, and pose drift. Log which information was actually consumed at the source selection branch, rather than relying only on launch arguments.

Falsification. If failures persist under oracle people but disappear under a changed low-level policy, the principal cause is likely execution rather than perception. If the accepted goals or actor exposure change between sensing modes, the comparison is confounded and needs redesign.

9.3 Gap 3: coordinated safety and liveness across planning layers

Priority: high; fit: direct; novelty confidence: moderate-to-low until a specific decision rule is defined. Freezing, cooperation, and hierarchical navigation are already studied [11, 22, 27, 45]. The opening is not the existence of a global/local hierarchy, but a verified policy for resolving disagreement between route intent, social planning, and safety supervision.

Hypothesis H3. Response-consistent handoff and event-driven route updates reduce prolonged veto loops and timeout probability without increasing contact or fall rates compared with fixed-rate replanning and independent supervision.

Contribution required. Define the state machine or optimization criterion governing yield, resume, detour, and route replacement. Measure delivered-route age, discarded or stale routes,

supervisor disagreement, recovery time, and progress after a yield. Match total CPU budget and prediction assumptions across alternatives.

Falsification. Higher route-update frequency without improved collision-free completion is an engineering change, not confirmation of the hypothesis. Likewise, reduced timeouts bought by more contacts do not establish a better trade-off unless the full frontier is reported.

9.4 Gap 4: full-body risk beyond planar clearance

Priority: medium; fit: requires extensions; novelty confidence: conditional. Whole-body control and three-dimensional avoidance already exist [13, 20, 47, 48]. A useful narrower question is when a planar swept footprint is inadequate for humanoid social navigation.

Hypothesis H4. A validated body-swept-volume approximation detects encounter risks missed by base clearance alone, especially during turning, lateral stepping, and recovery.

This requires link-level contact/clearance reference measurements and scenario designs that exercise torso/arm motion and stability. Start by quantifying the error of the current footprint abstraction. A more expensive full-body method is warranted only if that error affects decisions. Claims about contact injury or human safety would need a different validation scope than collision counting.

9.5 Gap 5: contextual comfort and nonreciprocal crowds

Priority: longer term; fit: requires human evaluation and richer actors. Social principles and contextual proxemics are well developed [17, 34]. The opportunity is to test whether response-aware motion creates behavior people actually find more predictable and comfortable, while avoiding an assumed universal radius.

Compare nonreactive, reciprocal, and data-driven pedestrian models first. Then evaluate selected robot behaviors with human judgments, controlling for speed, clearance, and morphology. Without this step, use the term *human-aware collision avoidance* or *proxemic-cost navigation* for the demonstrated capability rather than claiming broadly validated social intelligence.

10 A concrete experimental programme

10.1 Stage A: characterize the frozen locomotion interface

Collect command step, ramp, stop, turn, and combined-axis responses across initial speeds, directions, and repeated runs. Record requested command, normalized command, supervisor output, policy-consumed command, estimated state, evaluation state, and timestamps. Measure latency, gain, overshoot, stopping distance, lateral drift, and fall outcomes. Separate training, calibration, and held-out response sets by complete trajectories, not adjacent samples from one rollout.

Compare (i) instantaneous planar integration; (ii) current hand-set envelope; (iii) fitted deterministic response; and (iv) fitted response with residual uncertainty. Evaluate horizon-dependent error and interval coverage. Software slew limits are not automatically physical acceleration bounds. Calibrate the active profile and policy together; switching either changes the model.

10.2 Stage B: controlled oracle planning study

Freeze maps, source information, policy, robot geometry, actor trajectories, and goals. The repository's development methods—Nav2-only, reactive geometric, deterministic temporal, and full planner/supervisor—provide a starting matrix, but are not substitutes for published comparator implementations. Add a well-specified classical baseline, preferably DWA, VO/ORCA with explicit reciprocity settings, or TEB, and describe adaptation to the G1's command interface.

All methods must share unavoidable hardware/policy limits. If the same supervisor is applied to every method, state that the comparison concerns the upstream planner under a shared shield. If supervision itself is the independent variable, separate the experimental arms so that a shared filter does not erase the difference being measured. Match tuning budgets; select parameters on development scenarios only.

10.3 Stage C: factorial onboard robustness study

Use a small, interpretable factorial design before scaling to many methods. Suggested factors are response model, observation-age compensation, perception source, and forecast uncertainty treatment. Add localization/map substitutions as diagnostic blocks if the full factorial is too costly. Report interactions: for example, a response model may help only when track age is corrected.

Scenario families should include head-on passing, orthogonal crossing, overtaking, occlusion emergence, corridor blockage, narrow passage, moving crowds, and turn/stop transients. Vary crowd density, human-to-robot speed ratio, sensor dropout, map ambiguity, and actor reciprocity. Include people-free trials to isolate locomotion and route-following failures. Hold out at least one scene family to probe overfitting to the plaza.

Identical seeds are not sufficient for pairing live runs: wall-time scheduling and observation-dependent goal admission can change encounters. For causal replay, freeze actor trajectories and goals where feasible, record accepted events, and quantify residual nondeterminism. Treat nonreactive replay and interactive crowds as separate experimental regimes.

10.4 Stage D: metrics and statistical interpretation

Predeclare primary outcomes: complete mission without contact/fall, contact incidence, and algorithmic timeout. Secondary outcomes include minimum body clearance, contact count/severity proxy, time in specified proxemic regions, approach speed, stopping error, path/time efficiency, yield/recovery duration, supervisor interventions, and processing/observation/actuation latencies. State whether a contact metric is based on geometry, physics contacts, or a distance threshold.

Keep four denominators visible: planned trials, executed trials, valid trials, and successful trials. A collision is an algorithmic outcome; a missing required scoring stream is an experiment-validity problem. However, method-induced overload can itself cause missing data. Predeclaring failure classification and reporting invalidity by method prevents invalid-trial exclusions from hiding poor computational robustness.

Use confidence intervals for rates and paired scenario/seed differences where replay supports pairing. Bootstrap at the independent scenario/seed unit, with repetitions nested within it, not at individual telemetry samples. The repository's Wilson intervals and paired aggregation are useful starting points, but the statistical design still needs to specify the independent unit and missingness assumptions. For zero observed collisions in n independent trials, the one-sided

exact 95% upper bound is

$$p_{\text{upper}} = 1 - 0.05^{1/n}. \tag{4}$$

Even zero events in 100 independent trials leaves an upper bound near 3%. Repeated correlated scenarios do not justify the same interpretation. Choose the trial count from desired precision or a prespecified detectable effect, not from the first sequence of passing demonstrations.

Do not rank methods only by travel time on successful trials: this conditions on success and can reward a fast but unreliable method. Report completion and failure distributions alongside conditional efficiency. Avoid a single weighted score obscuring collisions, falls, invalidity, and timeouts.

10.5 Minimum publishable evidence package

1. A frozen protocol, policy/configuration hashes, source manifests, scenario definitions, and explicit information boundaries.
2. Held-out response-model error and calibration results.
3. Matched baselines plus ablations isolating response, delay, uncertainty, and supervision.
4. Multi-scenario, multi-seed mission results with retained failures, validity accounting, and uncertainty estimates.
5. Failure-case analysis linking sensor observations, decisions, applied commands, and physical outcomes.
6. Reproducible artifacts and a bounded contribution statement supported by those experiments.

11 Recommended positioning and conclusion

Suggested working title: *Observation-Age-Aware, Response-Calibrated Social Navigation for a Walking Humanoid: A Source-Audited Evaluation in MuJoCo and ROS 2.*

This title describes a proposed research direction, not completed results. The paper's eventual central claim should take a falsifiable form:

> Under a declared sensing, locomotion, and crowd model, the proposed calibration and decision procedure improves collision-free mission completion relative to matched baselines, and controlled ablations identify when the improvement depends on observation age and executable response.

Whether that statement becomes true is an experimental question. At present, the repository supports the platform and diagnosis portions of the work, while reliable onboard avoidance remains open.

The literature gives a clear boundary to novelty. Dynamics-aware avoidance, biped crowd planning, learned crowd interaction, sensor-driven G1 avoidance, cross-embodiment navigation, and physically grounded navigation benchmarks already exist. The most promising contribution is a carefully controlled account of their integration failure modes, together with a specific, validated mechanism that improves the safety–liveness trade-off. A rigorous negative result showing that a proposed response model does not help under realistic sensing would be more informative than an unqualified success claim based on selected oracle runs.

A Annotated evidence map of the 50 retained works

The entries below give each work's role in this review. They are not claims that every work received an equally detailed full-text review. Identity records and content-inspection levels are

listed separately in the audit.

Work	Contribution to the synthesis and transfer limitation
Kruse et al. [25]	Foundational human-aware navigation framing; supplies historical context rather than a contemporary G1 baseline.
Singamaneni et al. [42]	Faceted taxonomy of socially aware navigation; guards against reducing social behavior to clearance alone.
Learning-based review [2]	Recent comparative overview and benchmark perspective; motivates learned baselines without transferring its rankings to different sensing contracts.
Evaluation guidelines [17]	Defines multiple social-navigation principles and evaluation dimensions; supports separate safety and human-centered outcomes.
Nahum et al. [34]	Context-dependent proxemics taxonomy; argues against universal personal-space geometry as a complete comfort metric.
MuJoCo [44]	Physics-engine foundation; simulator choice alone is neither calibrated dynamics nor algorithmic novelty.
ROS 2 [31]	Middleware architecture and deployment context; application-level latency remains a measured property.
The Marathon 2 [30]	Nav2 architecture and navigation orchestration; not a G1 policy-response model.
SLAM Toolbox [29]	Mapping infrastructure; online mapping success must be separated from crowd avoidance.
DWA [16]	Dynamics-constrained velocity search baseline; makes generic acceleration-aware novelty claims inappropriate.
Velocity obstacles [15]	Relative-motion collision reasoning and admissible velocities; requires appropriate motion estimates and robot constraints.
ORCA [46]	Reciprocal collision-avoidance reference; actor reciprocity must match or be explicitly varied.
Sparse trajectory optimization [38]	Time/trajectory optimization comparator; the deployed controller model remains consequential.
Social force [21]	Interpretable pedestrian interaction model; not universal empirical human behavior.
Trautman et al. [45]	Dense-crowd cooperation perspective; motivates explicit freezing/progress metrics.
CADRL [12]	Learned decentralized collision avoidance; representation and training crowd assumptions need matching.
SARL [10]	Attention-based crowd interaction reasoning; useful learned-state baseline, separate from person detection.
RGL [11]	Graph relations and predictive planning; compare under the same sensing and execution model.
DS-RNN [28]	Temporal/spatial learning and partial observability; real wheeled transfer does not establish biped response equivalence.
SocNavBench [5]	Grounded, common evaluation scenarios and metrics; already occupies broad benchmark-novelty territory.
SocialGym 2.0 [9]	Configurable multi-agent navigation environments; robot interaction assumptions affect transfer.
JRDB [32]	Egocentric human perception dataset; relevant to misses/association, not a complete G1 closed-loop benchmark.
SCAND [24]	Human navigation demonstrations across robot morphologies; behavioral reference with different sensors and actuation.
DROW [4]	Learned sparse-range detection precedent; wheelchair-oriented original task differs from torso-mounted G1 person tracking.
DR-SPAAM [23]	Temporal/spatial person detection from range scans; projection and walking-motion domain shift require evaluation.
FAST-LIO2 [50]	Strong direct LiDAR–inertial estimation comparator; localization improvements alone do not establish navigation reliability.

Work	Contribution to the synthesis and transfer limitation
ORB-SLAM3 [8]	Visual/visual–inertial state-estimation alternative; requires a suitable camera pipeline.
Social LSTM [1]	Interaction-aware learned trajectory prediction; forecasting and decision quality are different outcomes.
Social GAN [19]	Diverse plausible futures; sample diversity does not by itself calibrate risk.
Trajectron++ [41]	Dynamical, heterogeneous, optionally ego-conditioned prediction; an important alternative to independent constant velocity.
Rudenko et al. [39]	Forecasting taxonomy, context, datasets, and metrics; motivates matching the prediction task to planning.
CBF-QP [3]	Formal safety filtering under assumptions; distinguishes invariant conditions from empirical rollout checks.
Safe learning survey [7]	Separates safe control, robustness, and certification; prevents overclaiming trained or heuristic safety.
Biped crowd navigation [6]	Direct historical prior art on biped safety in crowds; detailed experiments were not inspected because of access limitations.
RMA [26]	Rapid locomotion adaptation; adaptation competence is not a pedestrian-planning policy.
Massively parallel walking [40]	Efficient locomotion training; does not establish social mission completion.
Perceptive locomotion [33]	Robust exteroceptive/proprioceptive integration; primarily terrain competence rather than social acceptability.
ANYmal Parkour [22]	Capability-aware navigation hierarchy; demonstrates that embodiment-aware planning already exists.
Wheeled-legged autonomy [27]	Integrated mobility-aware navigation and long-range missions; different morphology but strong systems-level prior art.
ExBody [13]	Expressive humanoid control with locomotion; perceived comfort needs an additional evaluation.
DeepMimic [35]	Physics-based motion imitation foundation; animation/skill objectives differ from crowd safety.
Dynamics randomization [36]	Foundational sim-to-real robustness method; not a calibrated stopping certificate.
ASAP [20]	Residual action alignment and G1 transfer; full-text methods/tables examined, task is motion imitation rather than social navigation.
Safe humanoid locomotion [48]	Closest LiDAR/G1 safety/comfort competitor; inspected costs and tables support a qualified, scenario-dependent comparison.
FLUX [18]	Recent generative navigation and G1 transfer; inspected metric definitions and deployment timing constrain cross-paper comparisons.
OmniNavBench [43]	Physically embodied, multi-task benchmark; inspected protocol narrows the open benchmark contribution.
MetaUrban [49]	Compositional urban micromobility simulation; abstract-screened broader environment/embodiment coverage.
Habitat 3.0 [37]	Human/avatar collaboration and human-in-loop evaluation; abstract-screened scope differs from low-level G1 response.
DPed-VLN [14]	September 2026 dynamic-pedestrian language-navigation benchmark; abstract-only emerging evidence, not a direct Nav2 comparator.
Omni-Perception [47]	Raw-LiDAR omnidirectional legged avoidance; abstract-screened close competitor requiring deeper comparison before publication.

B Claim audit and immediate priorities

Potential claim	Assessment and required evidence
"First humanoid social-navigation system"	Contradicted by direct prior art; do not use.
"First dynamics-aware avoidance"	Contradicted by DWA, VO, biped planning, and subsequent embodied methods.
"Novel response-calibrated uncertainty-aware method"	Possible only after specifying the new mechanism and comparing the nearest methods under matched conditions.
"Sensor-only navigation implemented"	Supported by the documented information boundary and tests, within its stated graph/configuration scope.
"Reliable onboard avoidance achieved"	Not supported by the current documented collisions and timeouts.
"Hard real-time planning"	Not supported by a soft search budget or observed p95 runtime.
"Formal safety guarantee"	Requires valid model/disturbance assumptions and an appropriate proof; heuristic rollout filtering is insufficient.
"Socially comfortable motion"	Geometric proxies may be reported as proxies; perceived comfort requires human-centered evidence.
"Reproducible benchmark contribution"	Plausible with validated artifacts, controlled replay, external baselines, and new scientific findings; seeded development demos alone are insufficient.
"Simulation results transfer to hardware"	Requires hardware evidence; neither MuJoCo realism nor another paper's G1 deployment establishes this project's transfer.

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